KOHSUKE IDE

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RESEARCH INTERESTS

Visual world models, self-supervised visual learning, 3D vision, vision-language models, and diagnostic evaluation. I study how visual systems can learn predictive representations from time, geometry, and action, and how to evaluate whether those representations support reasoning and interaction beyond benchmark shortcuts.

EDUCATION

4/2025 – 3/2027 (expected) **M.Eng. in Computer Science** — **University of Tsukuba, Japan**
Advisor: **Prof. Yutaka Satoh**. GPA: **4.27/4.30**.

4/2021 – 3/2025 **B.Eng. in Applied Mechanics and Aerospace Engineering** — **Waseda University, Japan**
Advisor: **Prof. Kiyoshi Saito**. GPA: **3.29/4.00**. Minor in Computer Science. Undergraduate research on reinforcement learning for energy-system optimization.

SELECTED PUBLICATIONS AND MANUSCRIPTS

1. **Kohsuke Ide**, Ryousuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, Yuki M. Asano, and Yutaka Satoh.
Beyond Downstream Scores: Controlled Diagnostics for Point-Cloud Self-Supervised Learning.
Under review at **NeurIPS** 2026. Reviewer scores: **5/5/5**.

2. **Kohsuke Ide**, Ryousuke Yamada, Yue Qiu, Xianzheng Ma, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.
Beyond Single Object: Learning 3D Relations with Large Language Models.
IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (**CVPR Findings**), 2026.
project | paper | code

3. Ryousuke Yamada, **Kohsuke Ide**, Yoshihiro Fukuhara, Hirokatsu Kataoka, Gilles Puy, Andrei Bursuc, and Yuki M. Asano.
3D sans 3D Scans: Scalable Pre-training from Video-Generated Point Clouds.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
project | paper | code

4. **Kohsuke Ide**, Ryousuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.
Seeing Red, Thinking Bad: Color Bias in Vision Language Models.
International Conference on Pattern Recognition (**ICPR**), 2026.
project | paper | code

5. **Kohsuke Ide**, Ryousuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.
Colors You Can't See, Semantic Biases You Can't Ignore.
ICCV Workshop on Multimodal Reasoning and General Intelligence (**ICCV Workshop**), 2025.

RESEARCH EXPERIENCE

9/2025 – Present ***Research Assistant*, AIST** — **Tsukuba, Japan**

- Conduct research on visual world models, 3D understanding, vision-language models, and diagnostic evaluation.
- Lead projects from problem formulation and implementation through experiments and writing; this work has produced first-author papers at **CVPR Findings 2026** and **ICPR 2026**.
- Collaborate with researchers at Oxford's Visual Geometry Group on 3D and video representation learning.

1/2024 – 8/2025 ***Research Intern*, AIST** — **Tsukuba, Japan**

- Developed diagnostic evaluations for shortcut learning, language leakage, and visual biases in 3D and vision-language models.
- Built multi-object 3D reasoning benchmarks that require differences between scenes rather than language-only cues.
- Designed and implemented a video-to-point-cloud data-generation pipeline for scalable 3D pre-training.

RESEARCH EXPERIENCE (CONTINUED)

8/2024 – 3/2025 ***Research Intern*, Preferred Networks** **Tokyo, Japan**

- Developed and improved 3D reconstruction and free-viewpoint video pipelines.
- Optimized rendering and model components for efficient experimentation.

9/2021 – 3/2024 ***Machine Learning Engineer Intern*, Lightblue Technology** **Tokyo, Japan**

- Developed and deployed an object-tracking system for a work-efficiency monitoring application using Docker and AWS EC2.
- Improved human-action-recognition accuracy by 20% through feature engineering and Optuna-based hyperparameter search.

9/2023 – 10/2023 ***MLOps Engineer Intern*, M3** **Tokyo, Japan**

- Built **RecoverOOM**, an open-source, event-driven service for automatically recovering GKE CronJobs after out-of-memory failures.
- Implemented OOM detection through Cloud Logging and Pub/Sub, automatic memory-limit updates, and immediate job re-runs.

RESEARCH VISITS & INTERNATIONAL COLLABORATION

1/19 – 1/31/2026 **Research Visit** **University of Oxford, UK**
Research meetings with members of the Visual Geometry Group and Active Vision Laboratory, including Xianzheng Ma and Christian Rupprecht.

4/22 – 4/30/2026 **ASPIRE Computer Vision Workshop and Research Visit** **University of Oxford, UK**
Presented research and discussed follow-up work with researchers at Oxford VGG.

ACADEMIC SERVICE

- **Workshop Organizer**, **Visual General Intelligence: Vision Research Toward the AGI Era**, CVPR 2026.

HONORS & AWARDS

- **MIRU 2025 Interactive Presentation Award** (Jul 2025)
  Selected as one of 11 award recipients from 601 interactive presentations (top 1.8%) for "Can 3D Large Language Models Count to Three?"
- **WINC Hackathon Winner** (2023)
  Built a ResNet-based push-up counter optimized with TensorRT for Jetson Nano.

TECHNICAL SKILLS & LANGUAGES

- **Research:** Computer vision, visual world models, self-supervised learning, 3D reconstruction, vision-language models, diagnostic evaluation.
- **Programming and systems:** Python, C++, Go, PyTorch, CUDA, TensorRT, OpenCV, Git, Docker, Kubernetes, Linux, AWS, and GCP.
- **Languages:** Japanese (native); English (fluent; TOEFL iBT 118/120, TOEIC 990/990).